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# A Study of Accurate Correlation-Energy Prediction Through Various Complex ML Architectures

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## Abstract

Accurate and efficient calculation of the electronic correlation energy ( $E_{corr}$ ) is a major challenge in quantum chemistry: the accuracy “gold-standard”, CCSD(T), has a steep computational cost, whereas mean-field Hartree-Fock (HF) is efficient but lacks electron-electron correlation. In this report, we examine machine-learning (ML) models for the prediction of  $E_{corr}$  directly from HF-derived features. This approach aims to couple the efficiency of HF with a precise, numerically accurate measurement of  $E_{corr}$ . More precisely, we evaluate increasingly complex ML architectures: from a literature baseline model, that employs purely HF-derived descriptors through a multi-layer perceptron (MLP) to an edge-aware graph attention network (GAT) combining electronic, chemical and Smooth Overlap of Atomic Positions (SOAP) features. Our approach is explored on increasing complexity of datasets: from water molecules to alkanes up to seven atoms organic compounds. Results show the ML models approximate  $E_{corr}$  with near chemical accuracy, while preserving a competitive computational cost.

## 1 Introduction

In the theory of electronic structure methods,  $E_{corr}$ ’s describes the instantaneous Coulomb repulsion between electrons. Cheap methods such as the mean-field HF are not able compute any of it. Although methods such as CCSD(T) capture almost exactly this correlation, the cost scaling with system size is a severe impediment<sup>4,9</sup>. Recent work has explored strategies that anchors the model on HF and targets  $E_{corr}$ . Within this strategy framework, early efforts demonstrated that lower-level features could predict  $E_{corr}$ , but these studies were often limited in the chemical space considered or in the richness of molecular representations<sup>6,8</sup>. The closest predecessor to our work is DeePHF, which predicts the correlation energy from symmetry-projected local HF density matrices and achieved state-of-the-art accuracy on organic benchmarks<sup>2</sup>. Yet DeePHF relies exclusively on orbital-derived (electronic) descriptors with no explicit bonding network context. The original paper of DeePHF mentioned that adding atomic descriptors would hinder transferability, but makes no claim on potential in-domain accuracy improvements of adding these descriptors. Our main question is: does orbital representation alone capture all the information relevant to  $E_{corr}$  or can the local chemical environment improve performance by providing a complementary signal that orbital features fail to capture? This work therefore explores the coupling of electronic (HF orbitals) and explicit geometric descriptors (per-atom SOAP for local environment, molecule identity-based atom and bond features) for the estimation of  $E_{corr}$  from  $E_{HF}$ , studying if their combination improves upon the baseline set by the orbital-only paradigm.

To facilitate accessibility and reproducibility, we provide an application interface for interactive use of the trained models.

## 2 Data

### 2.1 Basic Data Type and Datasets

The study is centered on molecular systems represented with their atomic properties and HF electronic structures. The target quantity is the electronic correlation energy<sup>7</sup>:

$$E_{corr} = E_{exact} - E_{HF} \quad (1)$$

where  $E_{exact}$  is the energy of highest theory level of post-HF available (CCSD(T) or MP2 depending on the dataset) and  $E_{HF}$  is the HF energy in the cc-PVDZ basis set. The computation of  $E_{corr}$  with these methods becomes intractable for large molecules<sup>4,10</sup>. To evaluate the performance of our method with molecular complexity, the experiments were conducted on successive ablation of the MOB-ML benchmark dataset<sup>3</sup>. This was done by increasing molecular size and diversity: water serving as a low-complexity reference case and sanity check, short alkanes with few heavy atoms, introducing mild structural variations and QM7b-T molecules with up to 7 heavy atoms which has a much broader chemical space. GDB-13-B allows to test if our method scales to larger molecules.

### 2.2 Attributes, descriptors, and processing

Each atom carries one or more of three per-atom feature blocks; bonded atom pairs carry an additional per-edge block (Table 1). The four descriptor variants compared in §4.1 are progressive concatenations of the per-atom blocks: `chemical` (17), `chemical_elec` (125), `chemical_soap` (1529), `chemical_elec_soap` (1637). Preprocessing is uniform across all 68 cells (Table 2); no data augmentation is performed, as rotational invariance is enforced by descriptor construction (eigenvalues of density-matrix blocks; SOAP power-spectrum invariants).

| Block                      | Scope    | Dim      | Source            | Captures                                                                                                                        |
|----------------------------|----------|----------|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <code>chemical</code>      | per-atom | 17       | RDKit             | element one hot (6), atomic number, hybridization one hot (7), aromaticity binary, formal charge, implicit H, electronegativity |
| <code>electronic</code>    | per-atom | 108      | PySCF(HF/cc-pVTZ) | eigenvalues of locally-projected HF density matrix <sup>2</sup>                                                                 |
| <code>SOAP</code>          | per-atom | 1512     | DScribe           | 3D local environment ( $r_{cut}=6$ Å, $n_{max}=8$ , $l_{max}=6$ ) <sup>1</sup>                                                  |
| <code>edge features</code> | per-edge | 5 (+ 16) | RDKit + geometry  | bond-type one-hot (4) + Euclidean distance (+ RBF distance (16 Gaussians on $[0, 10]$ Å))                                       |

Table 1: Feature blocks. Construction follows DeePHF: 12 difference-of-Gaussian radials  $\times (s, p, d) = 108$  basis functions per atom; per- $(n, l)$  block eigenvalues yield rotational invariance.

| Step                    | Operation                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------|
| Graph construction      | openbabel converts XYZ to SDF files, processed then by RDKit for feature extraction                     |
| Per-block normalization | each per-atom block independently z-scored on the training split; same transform applied to val/test    |
| Element bias            | per-element OLS fit on training targets; frozen and added to the NN output as a non-trainable embedding |
| Split                   | deterministic 80/10/10 train/val/test, identical across all 68 cells                                    |

Table 2: Preprocessing pipeline.

## 3 Methods

### 3.1 Multi-Layer Perceptron

Following Chen et al.<sup>2</sup>’s architecture, we use a per-atom MLP (which we will refer to as *DeePHF-MLP*). Each atom  $i$  is described by a feature vector  $\mathbf{x}_i$ : a shared MLP predicts an atomic contribution  $\epsilon_i = f_{\theta}(\mathbf{x}_i)$ , and the molecular correlation energy is

$$E_{corr} = \sum_i \epsilon_i. \quad (2)$$

This is a useful sanity check: it tells us how far we can get from per-atom features alone, without modeling any structure. But by construction it cannot see which atoms are bonded, how far apart they are, or how electronic correlation depends on the broader molecular geometry. Since  $E_{\text{corr}}$  comes from coupled electron motion across the whole molecule, we expect this to matter. To capture it, we need a model that operates over the molecular graph.

### 3.2 Graph Convolutional Networks

We turn to graph convolutional networks (GCNs). A molecule becomes a graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  with atoms as nodes and bonds as edges. At each layer, every atom updates its hidden state by aggregating messages from its neighbours<sup>5</sup>:

$$\mathbf{h}_i^{(\ell+1)} = \text{update} \left( \mathbf{h}_i^{(\ell)}, \bigoplus_{j \in \mathcal{N}(i)} \phi(\mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}) \right), \quad (3)$$

where  $\mathbf{e}_{ij}$  are edge features and  $\bigoplus$  is a permutation-invariant aggregator (summation in our case). The message function was used as in Gilmer et al.<sup>5</sup>, and an MLP with GELU activation was used for the updates. After  $T$  layers, each atom’s representation carries information from its  $T$ -hop neighbourhood. The readout employs an MLP such that:  $E_{\text{corr}} = \sum_i g_\theta(\mathbf{h}_i^{(T)})$ .

### 3.3 Graph Attention Networks (GATs)

A GAT computes per-neighbour weights

$$\alpha_{ij} = \text{softmax}_{j \in \mathcal{N}(i)} a(\mathbf{W}\mathbf{h}_i, \mathbf{W}\mathbf{h}_j), \quad (4)$$

and aggregates as  $\sum_j \alpha_{ij} \mathbf{W}\mathbf{h}_j$ . The model can now weight neighbours adaptively.

### 3.4 Edge-aware attention function design

We fix this with an additive edge bias. Following the multi-head GAT formulation<sup>11</sup> with an added edge channel, the attention logit becomes

$$\alpha_{ij} = \text{softmax}_{j \in \mathcal{N}(i)} \text{LeakyReLU}[\mathbf{a}_{\text{dst}}^\top(\mathbf{W}\mathbf{h}_i) + \mathbf{a}_{\text{src}}^\top(\mathbf{W}\mathbf{h}_j) + g(\mathbf{e}_{ij})], \quad (5)$$

where  $\mathbf{W} \in R^{d \times d}$  is a shared linear projection,  $\mathbf{a}_{\text{src}}, \mathbf{a}_{\text{dst}} \in R^{d/H}$  are per-head learned scalar projections, and  $g(\mathbf{e}_{ij}) = \mathbf{a}_{\text{edge}}^\top(\mathbf{W}_e \mathbf{e}_{ij})$  is a scalar bias computed from edge features (bond type, RBF-expanded distance). The aggregation then becomes  $\mathbf{m}_i = \sum_j \alpha_{ij} \mathbf{W}\mathbf{h}_j$ , reusing  $\mathbf{W}$  rather than introducing a separate value matrix. The intuition is simple: bonded pairs, near non-bonded pairs, and distant non-bonded pairs should be able to receive distinct attention weights even when their endpoints’ node states alone would have scored them identically.

The intuition is simple: bonded pairs, distant non-bonded pairs, and geometrically equivalent atoms should be able to receive distinct attention weights regardless of how their node states alone would have scored them.

## 4 Results

Each cell is trained and evaluated in-domain on its own MOB-ML subset under the protocol of §3; the only sources of variation are architecture and feature richness. All 16 architecture  $\times$  descriptor combinations are evaluated per dataset, plus one `bias_only` OLS baseline (68 cells total).

### 4.1 In-domain accuracy across the $4 \times 4 \times 4$ grid

Figure 1 reports test MAE on all 68 cells; Table 3 extracts the best cell per dataset.

Three of four datasets show the same pattern: the edge-aware attention architecture combined with the full descriptor set yields the lowest MAE, reaching chemical accuracy by margins of  $250\times$  (water),  $26\times$  (alkanes), and missing it by 12% on GDB-13-T. QM7b-T is the exception:

| Dataset          | $n_{\text{train}}$ | MAE ( $\text{mE}_h$ ) | Architecture | Descriptor         |
|------------------|--------------------|-----------------------|--------------|--------------------|
| H <sub>2</sub> O | 800                | 0.0063                | D_gat_edge   | chemical_elec_soap |
| Alkanes          | 1843               | 0.0622                | D_gat_edge   | chemical_elec_soap |
| QM7b-T           | 3534               | 0.9597                | A_mlp        | chemical_elec      |
| GDB-13-T         | 4752               | 1.7836                | D_gat_edge   | chemical_elec_soap |

Table 3: Best-performing architecture/descriptor combination per dataset (seed 43, in-domain test MAE). Chemical accuracy:  $1.59 \text{ mE}_h$ .

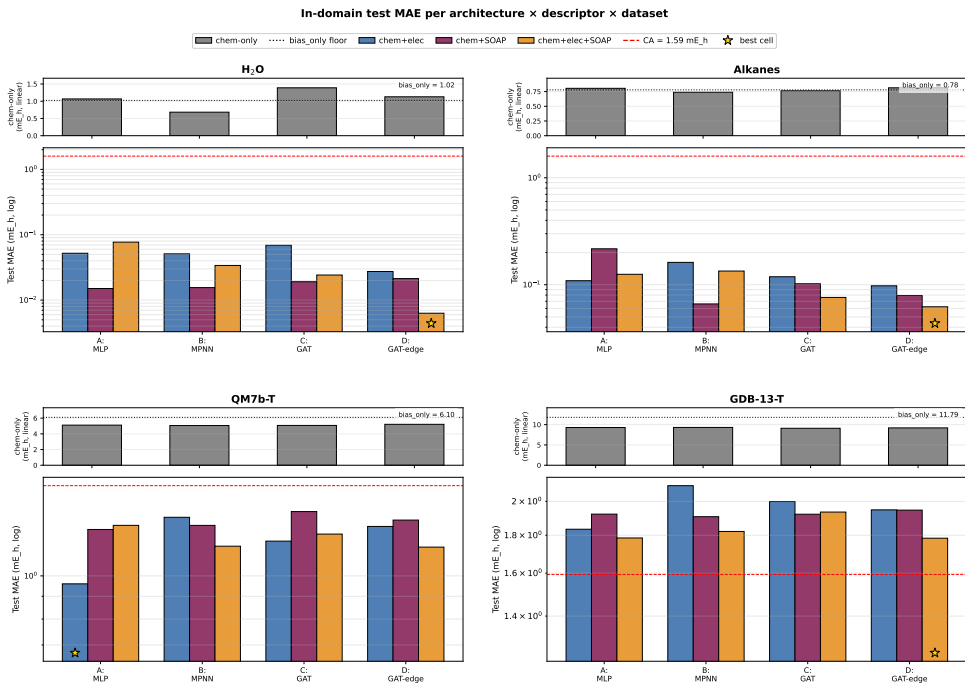


Figure 1: Test MAE per architecture  $\times$  descriptor  $\times$  dataset. Top strip (linear): chemical-only and bias-only baselines. Bottom (log): the three richer descriptor variants, with chemical accuracy ( $1.59 \text{ mE}_h$ ) as a red dashed line. Gold star: best cell per dataset.

the simplest per-atom MLP with the simplest informative descriptor wins at  $0.96 \text{ mE}_h$ , and the richer `chemical_elec_soap` variant fails to benefit any architecture. Two observations support the architecture–descriptor interaction reading. First, where the richest descriptors win (three of four datasets), the most expressive architecture wins with them. Second, where the simpler architecture wins (QM7b-T), it wins specifically with the simpler descriptor; SOAP is not just unused but actively unhelpful in that regime.

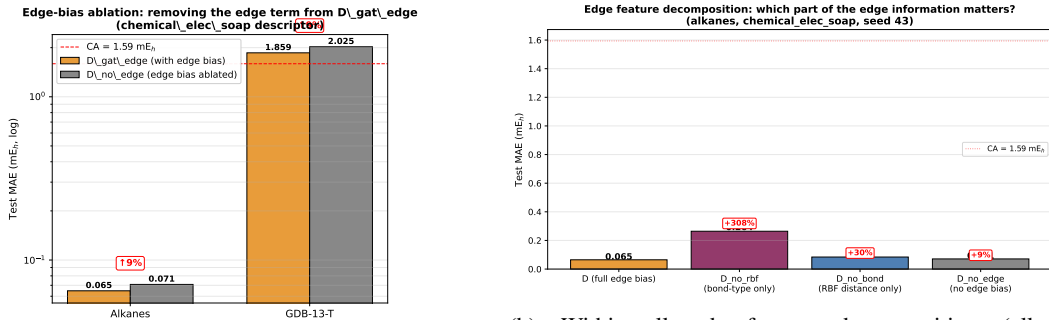
As a single-number summary: averaged across datasets, the relative MAE reduction from `chemical` to `chemical_elec_soap` is A 83.2%, B 83.6%, C 85.6%, D 87.5% (mean-feature scaling slope), monotonic with architectural inductive bias, with the full A to D spread of 4.3 pp concentrated at the two qualitative transitions: A $\rightarrow$ B (aggregation, +0.4 pp, within per-dataset noise) is negligible, while B $\rightarrow$ C (attention, +2.0 pp) and C $\rightarrow$ D (edge bias, +1.9 pp) carry the effect. The differentiation is thus at the introduction of attention and again at the geometric edge channel, not at message passing alone. The 4.3 pp end-to-end magnitude is modest; the substantive evidence for the architecture–descriptor interaction lies in the targeted ablations of §4.2.

## 4.2 Mechanistic evidence: locating the lever

The grid in §4.1 raises an architectural question: why would a geometry-aware architecture extract more from richer descriptors than a per-atom MLP? Correlation energy is a pairwise quantity: it arises from coupled electron motion between atoms, and the coupling decays with inter-atom distance. A per-atom MLP sees each atom in isolation and can never express such pairwise distance-

modulated interactions, no matter how rich its descriptor. The edge-aware GAT can, through exactly one component: the edge-bias term  $b(e_{ij})$  in Eq. 5, which injects continuous distance (via the RBF channel and the bond-length channel) as well as bond connectivity (via the bond-type channel) into the attention logit. If  $b(e_{ij})$  is the mechanism, removing it should degrade performance specifically on descriptors with geometric content, since the edge bias is the network’s only path for injecting pairwise distance into message weighting.

We test this at two scales (Figure 2). (a) Across all 16 (dataset  $\times$  descriptor) cells of architecture D, we re-train with and without  $b(e_{ij})$  (D\_gat\_edge vs D\_no\_edge — the latter structurally identical to architecture C, same parameter budget). (b) Within the alkanes/chemical\_elec\_soap/D cell, we further decompose by independently zeroing the 16-dim RBF distance sub-channel and the distance channel (D\_no\_rbf, bond-type only) and the 4-dim bond-type (D\_no\_bond, RBF+distance only).



(a) Cross-cell edge-bias ablation.

(b) Within-cell edge-feature decomposition (alkanes/chemical\_elec\_soap/D, seed 43).

Figure 2: Locating the mechanism. (a) MAE with  $g(e_{ij})$  compared to MAE with it intact, across (alkanes, GDB 13-T) of architecture D to assess the edge bias helps most on descriptors with geometric content. (b) Within the edge bias, the RBF distance channel carries the lever: zeroing it (D\_no\_rbf) is four times worse than zeroing the entire edge bias, meaning categorical bond-type without distance is actively misleading.

Both panels point at the same lever from different angles. In (a), the edge-bias term contributes consistently across chemistries at the full descriptor: zeroing it costs 9% on alkanes and 8.9% on GDB-13-T, confirming the mechanism is not dataset-specific. In (b), zeroing the continuous RBF channel costs 308% while zeroing the categorical bond-type costs only 30%, identifying distance as the load-bearing sub-channel. The non-obvious finding is that D\_no\_rbf is dramatically worse than D\_no\_edge: bond-type information without distance grounding actively misleads attention, more harmful than no edge bias at all. The full D\_no\_edge ablation costs only 9% on this alkanes cell, consistent with the 1512-dim per-atom SOAP encoding already carrying most local geometric information; the edge-bias’s role is to inject pairwise distance into the attention scores, complementing rather than duplicating SOAP’s per-atom geometric encoding.

## 5 Conclusion

We tested whether SOAP geometric descriptors improve  $E_{\text{corr}}$  prediction beyond DeePHF’s electronic descriptors when allied to basic molecular chemical properties, across four MOB-ML subsets and four architectures of increasing inductive bias. The combination of an edge-aware graph attention architecture with the full descriptor set (chemical\_elec\_soap) achieves chemical accuracy on three of four datasets (H<sub>2</sub>O: 0.006 mE<sub>h</sub>; alkanes: 0.062 mE<sub>h</sub>; QM7b-T via the simpler A / chemical\_elec cell: 0.960 mE<sub>h</sub>), and approaches it on GDB-13-T (1.78 mE<sub>h</sub>, 12% above chemical accuracy). The stated hypothesis that architectural inductive bias amplifies the value of richer descriptors is confirmed on the mean feature-scaling slope (D 87.5% > C 85.6% > B 83.6% > A 83.2%). In other words, we conclude that geometric features can be a strong asset in post-HF method computation, but they require an equally powerful and expressive architecture to leverage their capacity and representational power.

Future work includes broader seed coverage to tighten the variance estimates and integrating the trained models into a geometry-optimization workflow that uses the predicted  $E_{\text{corr}}$  in place of post-HF reference calculations.

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